

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **GPA:** 3.5 / 4.0

- **Relevant Coursework:** Linear Algebra, Discrete Math, OOP, Computer Organization, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 – Present

East Lansing, MI

- Replaced multi-day manual onboarding with one automated flow via GitHub + Slack APIs so contributors collaborate in the right software repositories and channels in under 30 seconds, cutting setup time an estimated 97% across more than 100 members.
- Retired paper attendance by issuing short-lived QR tokens cached at the edge with S3 and CloudFront, validating check-in in real time and sustaining 150+ scans per event without redundant authentication round-trips.
- Hardened how the club ships software with GitHub Actions coverage gates above 90%, branch protection on production merges, and review standards that reduced escaped defects by an estimated 65%.

PROJECTS

Dadei — Ambient AI Assistant | *Python, FastAPI, PostgreSQL, pgvector, Redis, RQ, Gemini API, faster-whisper, pyannote-audio, PyTorch, Docker*

May 2025 – Nov. 2025

dadei.app

- Shipped an ambient assistant where each conversation close runs an agentic AI agent with tool-calling on Gemini 2.5 Pro, proposing episodic memories and calendar-style actions with evidence quotes tied to a transcript manifest instead of brittle wake-word-only UX.
- Ran local speech understanding on CPU using **faster-whisper** (**large-v3-turbo**, int8) for word-timed transcripts and machine learning models for diarization (**pyannote/speaker-diarization-3.1**) with an online centroid plus capped prototype bank and cosine gates so diarization-attributed turns stay stable across sessions.
- Implemented hybrid retrieval over **pgvector** (Gemini **gemini-embedding-001**, IVFFlat cosine ANN) with semantic-lexical fusion, 30-day recency decay, and optional RRF before prompting—tightening context for validation-heavy extraction and supporting data analysis when tuning fusion weights.
- Delivered integrated software pipelines under Docker with optional Redis/RQ workers, repeatable command-line builds, and CI that boots a real **pgvector** Postgres service for pytest—automated testing and validation alongside bounded duplicate delivery for 100+ registered personal networks after Redis dedupe claims.

WizViz — SpartaHack X | *MediaPipe, OpenCV, TensorFlow, NumPy*

devpost.com/software/wizviz

- Won first place in the Interactive Media track at SpartaHack X with a gesture-driven wizard duel where live body pose becomes spell casting, blocking, and ability triggers against an on-device AI opponent—no game engine, no cloud round-trips.
- Mapped full-body pose to gameplay by extracting 33 skeletal landmarks each frame, classifying them into discrete actions, and tightening the inference loop until the experience held 60 FPS with sub-20ms input lag.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Golang, C++, Bash

ML & AI LLMs, agentic AI, PyTorch, faster-whisper, pyannote-audio, Gemini API, pgvector, TensorFlow, MediaPipe, spaCy

Frameworks FastAPI, Electron, Vite, React, Redis, RQ, Docker, GitHub

Databases & Infrastructure PostgreSQL, GitHub Actions, AWS (S3, CloudFront)